

量子力学 (A) (2026 年春)

第 3 章作业

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1

设粒子限制在矩形匣子中运动, 即

$$V(x, y, z) = \begin{cases} 0, & 0 < x < a, 0 < y < b, 0 < z < c, \\ \infty, & \text{其余区域.} \end{cases}$$

求粒子的能量本征值和本征波函数。如 $a = b = c$, 讨论能级的简并度。

解: 在势阱内部 $0 < x < a, 0 < y < b, 0 < z < c$ 时, $V(x, y, z) = 0$; 在边界及阱外, $V = \infty$, 因此波函数必须满足边界条件

$$\psi(0, y, z) = \psi(a, y, z) = 0,$$

$$\psi(x, 0, z) = \psi(x, b, z) = 0,$$

$$\psi(x, y, 0) = \psi(x, y, c) = 0.$$

在阱内, 定态薛定谔方程为

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(x, y, z) = E \psi(x, y, z),$$

即

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2mE}{\hbar^2} \psi = 0.$$

令波函数分离变量:

$$\psi(x, y, z) = X(x)Y(y)Z(z).$$

代入方程并除以 XYZ , 得

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} + \frac{Z''(z)}{Z(z)} = -\frac{2mE}{\hbar^2}.$$

由于三项分别只依赖于不同变量, 故可分别令

$$\frac{X''}{X} = -k_x^2, \quad \frac{Y''}{Y} = -k_y^2, \quad \frac{Z''}{Z} = -k_z^2,$$

并有

$$k_x^2 + k_y^2 + k_z^2 = \frac{2mE}{\hbar^2}.$$

于是三个方向分别满足

$$X'' + k_x^2 X = 0, \quad Y'' + k_y^2 Y = 0, \quad Z'' + k_z^2 Z = 0.$$

结合边界条件

$$X(0) = X(a) = 0, \quad Y(0) = Y(b) = 0, \quad Z(0) = Z(c) = 0,$$

可知只允许正弦形式的非平凡解:

$$\begin{aligned} X_{n_x}(x) &= A_x \sin\left(\frac{n_x \pi x}{a}\right), \\ Y_{n_y}(y) &= A_y \sin\left(\frac{n_y \pi y}{b}\right), \\ Z_{n_z}(z) &= A_z \sin\left(\frac{n_z \pi z}{c}\right), \end{aligned}$$

其中

$$n_x, n_y, n_z = 1, 2, 3, \dots$$

且

$$k_x = \frac{n_x \pi}{a}, \quad k_y = \frac{n_y \pi}{b}, \quad k_z = \frac{n_z \pi}{c}.$$

因此能量本征值为

$$E_{n_x, n_y, n_z} = \frac{\hbar^2}{2m} \left[\left(\frac{n_x \pi}{a}\right)^2 + \left(\frac{n_y \pi}{b}\right)^2 + \left(\frac{n_z \pi}{c}\right)^2 \right] = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right).$$

再由归一化条件

$$\int_0^a |X_{n_x}(x)|^2 dx = 1, \quad \int_0^b |Y_{n_y}(y)|^2 dy = 1, \quad \int_0^c |Z_{n_z}(z)|^2 dz = 1,$$

并利用

$$\int_0^a \sin^2\left(\frac{n_x \pi x}{a}\right) dx = \frac{a}{2},$$

同理对 y, z 方向也成立, 可得

$$A_x = \sqrt{\frac{2}{a}}, \quad A_y = \sqrt{\frac{2}{b}}, \quad A_z = \sqrt{\frac{2}{c}}.$$

故归一化本征波函数为

$$\psi_{n_x n_y n_z}(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right),$$

其中 $0 < x < a, 0 < y < b, 0 < z < c$; 阱外 $\psi = 0$ 。

当 $a = b = c \equiv L$ 时, 能量退化为

$$E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2).$$

此时能量仅依赖于 $n_x^2 + n_y^2 + n_z^2$, 因此凡是满足相同平方和的不同三元组 (n_x, n_y, n_z) 都对应同一能级, 从而出现简并。

简并度的讨论如下:

- 若 n_x, n_y, n_z 三个量两两不同, 例如 $(1, 2, 3)$, 则其所有排列对应 6 个不同本征态, 因此该能级为 6 重简并;
- 若其中两个相同、一个不同, 例如 $(1, 1, 2)$, 则不同排列有 3 个, 因此该能级为 3 重简并;
- 若三个都相同, 例如 $(1, 1, 1)$, 则只有 1 个本征态, 因此该能级非简并。

此外, 还可能出现偶然简并, 即并非简单排列关系的不同三元组也满足相同的平方和; 这时总简并度应等于所有满足

$$n_x^2 + n_y^2 + n_z^2 = \text{常数}$$

的正整数三元组个数。

2

设粒子处于一维无限深方势阱中,

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x < 0, x > a. \end{cases}$$

证明处于能量本征态 $\psi_n(x)$ 的粒子,

$$\bar{x} = a/2, \quad \overline{(x - \bar{x})^2} = \frac{a^2}{12} \left(1 - \frac{6}{n^2 \pi^2}\right).$$

讨论 $n \rightarrow \infty$ 的情况, 并与经典力学计算结果比较。

证明：一维无限深方势阱的归一化能量本征函数为

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, 3, \dots,$$

阱外 $\psi_n(x) = 0$, 故

$$|\psi_n(x)|^2 = \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right).$$

先计算 $\bar{x}$:

$$\begin{aligned} \bar{x} &= \langle x \rangle \\ &= \int_0^a x |\psi_n(x)|^2 dx \\ &= \frac{2}{a} \int_0^a x \sin^2\left(\frac{n\pi x}{a}\right) dx \\ &= \frac{1}{a} \int_0^a x \left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx \\ &= \frac{1}{a} \int_0^a x dx - \frac{1}{a} \int_0^a x \cos\left(\frac{2n\pi x}{a}\right) dx \\ &= \frac{1}{a} \left. \frac{x^2}{2} \right|_0^a - \frac{1}{a} \left. \frac{x \sin\left(\frac{2n\pi x}{a}\right)}{\frac{2n\pi}{a}} + \frac{\cos\left(\frac{2n\pi x}{a}\right)}{\left(\frac{2n\pi}{a}\right)^2} \right|_0^a \\ &= \frac{a}{2}. \end{aligned}$$

对于方差, 先计算 $\langle x^2 \rangle$:

$$\begin{aligned} \langle x^2 \rangle &= \int_0^a x^2 |\psi_n(x)|^2 dx \\ &= \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx \\ &= \frac{1}{a} \int_0^a x^2 \left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx \\ &= \frac{1}{a} \int_0^a x^2 dx - \frac{1}{a} \int_0^a x^2 \cos\left(\frac{2n\pi x}{a}\right) dx \\ &= \frac{1}{a} \left. \frac{x^3}{3} \right|_0^a - \frac{1}{a} \left. \frac{x^2 \sin\left(\frac{2n\pi x}{a}\right)}{\frac{2n\pi}{a}} + \frac{2x \cos\left(\frac{2n\pi x}{a}\right)}{\left(\frac{2n\pi}{a}\right)^2} - \frac{2 \sin\left(\frac{2n\pi x}{a}\right)}{\left(\frac{2n\pi}{a}\right)^3} \right|_0^a \\ &= \frac{a^2}{3} - \frac{a^2}{2n^2 \pi^2}. \end{aligned}$$

因此:

$$\begin{aligned} \overline{(x - \bar{x})^2} &= \langle x^2 \rangle - \langle x \rangle^2 \\ &= \left(\frac{a^2}{3} - \frac{a^2}{2n^2 \pi^2} \right) - \left(\frac{a}{2} \right)^2 \\ &= \frac{a^2}{12} \left(1 - \frac{6}{n^2 \pi^2} \right). \end{aligned}$$

当 $n \rightarrow \infty$ 时,

$$\overline{(x - \bar{x})^2} \rightarrow \frac{a^2}{12}.$$

经典情形下, 区域 $(0, a)$ 中的粒子被击中的概率密度是 $1/a$, 故

$$\langle x \rangle_{\text{cl}} = \frac{1}{a} \int_0^a x \, dx = \frac{a}{2},$$

$$\langle x^2 \rangle_{\text{cl}} = \frac{1}{a} \int_0^a x^2 \, dx = \frac{a^2}{3},$$

所以

$$\overline{(x - \bar{x})^2}_{\text{cl}} = \langle x^2 \rangle_{\text{cl}} - \langle x \rangle_{\text{cl}}^2 = \frac{a^2}{12}.$$

因此当 $n \rightarrow \infty$ 时, 量子结果趋于经典结果。

3

设粒子处于一维无限深方势阱中,

$$V(x) = \begin{cases} 0, & |x| < a/2, \\ \infty, & |x| > a/2. \end{cases}$$

处于基态 ($n = 1$, 见 2.2 节式 (12)). 求粒子的动量分布。

解: 基态波函数为

$$\psi_1(x) = \begin{cases} \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right), & |x| < \frac{a}{2}, \\ 0, & |x| > \frac{a}{2}. \end{cases}$$

动量表象波函数为

$$\varphi_1(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \psi_1(x) e^{-ipx/\hbar} \, dx.$$

因此

$$\begin{aligned} \varphi_1(p) &= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{a}} \int_{-a/2}^{a/2} \cos\left(\frac{\pi x}{a}\right) e^{-ipx/\hbar} \, dx \\ &= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{a}} \int_{-a/2}^{a/2} \cos\left(\frac{\pi x}{a}\right) \cos\left(\frac{px}{\hbar}\right) \, dx \\ &= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{a}} \cdot \frac{1}{2} \int_{-a/2}^{a/2} \left[\cos\left(\left(\frac{\pi}{a} - \frac{p}{\hbar}\right)x\right) + \cos\left(\left(\frac{\pi}{a} + \frac{p}{\hbar}\right)x\right) \right] \, dx \\ &= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{a}} \left[\frac{\sin\left[\left(\frac{\pi}{a} - \frac{p}{\hbar}\right)\frac{a}{2}\right]}{\frac{\pi}{a} - \frac{p}{\hbar}} + \frac{\sin\left[\left(\frac{\pi}{a} + \frac{p}{\hbar}\right)\frac{a}{2}\right]}{\frac{\pi}{a} + \frac{p}{\hbar}} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{a}} \cos\left(\frac{p}{\hbar}a\right) \left[\frac{1}{\frac{\pi}{a} - \frac{p}{\hbar}} + \frac{1}{\frac{\pi}{a} + \frac{p}{\hbar}} \right] \\
&= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{a}} \cos\left(\frac{p}{\hbar}a\right) \frac{2\pi/a}{\left(\frac{\pi}{a}\right)^2 - \frac{p^2}{\hbar^2}} \\
&= \sqrt{\frac{a}{\pi\hbar}} \frac{2\pi \cos\left(\frac{pa}{\hbar}\right)}{\pi^2 - \left(\frac{pa}{\hbar}\right)^2}.
\end{aligned}$$

故动量分布为

$$w(p) = |\varphi_1(p)|^2 = \frac{4a}{\pi\hbar} \frac{\pi^2 \cos^2\left(\frac{pa}{\hbar}\right)}{\left[\pi^2 - \left(\frac{pa}{\hbar}\right)^2\right]^2}.$$

即

$$w(p) dp = \frac{4a}{\pi\hbar} \frac{\pi^2 \cos^2\left(\frac{pa}{\hbar}\right)}{\left[\pi^2 - \left(\frac{pa}{\hbar}\right)^2\right]^2} dp$$

表示粒子动量落在 $(p, p + dp)$ 内的概率。

4

设粒子处于无限深方势阱

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x < 0, x > a \end{cases}$$

中, 粒子波函数为 $\psi(x) = Ax(x-a)$, A 为归一化常数。

(a) 求 A ;

(b) 求测得粒子处于能量本征态

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

的概率 P_n , 特别是 P_1 。

提示: 用 $\psi_n(x)$ 展开, $\psi(x) = \sum_n C_n \psi_n(x)$, $P_n = |C_n|^2$ 。

(c) 作图, 比较 $\psi(x)$ 与 $\psi_1(x)$ 曲线。从 $P_1 \gg P_n (n \neq 1)$ 来说明两条曲线非常相似, 即 $\psi(x)$ 几乎与基态 $\psi_1(x)$ 完全相同。

解:

(a) 由归一化条件

$$\begin{aligned}
 1 &= \int_0^a |\psi(x)|^2 dx \\
 &= A^2 \int_0^a x^2(x-a)^2 dx \\
 &= A^2 \int_0^a (x^4 - 2ax^3 + a^2x^2) dx \\
 &= A^2 \left(\frac{x^5}{5} - \frac{ax^4}{2} + \frac{a^2x^3}{3} \right) \Big|_0^a \\
 &= A^2 a^5 \left(\frac{1}{5} - \frac{1}{2} + \frac{1}{3} \right) \\
 &= \frac{A^2 a^5}{30}.
 \end{aligned}$$

故

$$A = \sqrt{\frac{30}{a^5}}.$$

(b) 将 $\psi(x)$ 按本征函数展开:

$$\psi(x) = \sum_{n=1}^{\infty} C_n \psi_n(x), \quad C_n = \int_0^a \psi_n(x) \psi(x) dx.$$

$$C_n = \sqrt{\frac{2}{a}} A \int_0^a x(x-a) \sin\left(\frac{n\pi x}{a}\right) dx.$$

其中

$$\begin{aligned}
 \int_0^a x(x-a) \sin\left(\frac{n\pi x}{a}\right) dx &= \int_0^a (x^2 - ax) \sin\left(\frac{n\pi x}{a}\right) dx \\
 &= \frac{2a^3}{n^3 \pi^3} [(-1)^n - 1].
 \end{aligned}$$

故

$$C_n = \sqrt{\frac{2}{a}} A \cdot \frac{2a^3}{n^3 \pi^3} [(-1)^n - 1] = \frac{4\sqrt{15}}{n^3 \pi^3} [(-1)^n - 1].$$

$$P_n = |C_n|^2 = \frac{240}{n^6 \pi^6} [(-1)^n - 1]^2 = \begin{cases} 0, & n \text{ 为偶数,} \\ \frac{960}{n^6 \pi^6}, & n \text{ 为奇数.} \end{cases}$$

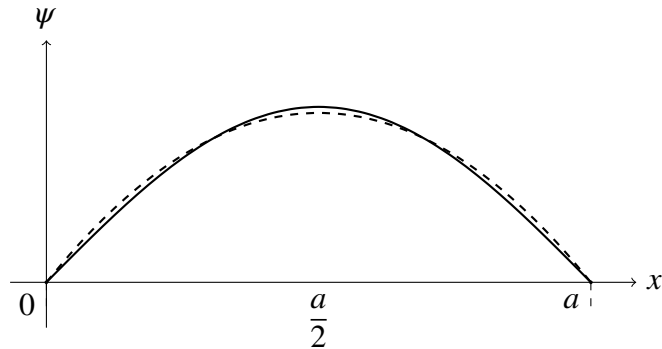
$$\Rightarrow P_1 = \frac{960}{\pi^6}.$$

注意到 $P_1 \approx 0.999 \approx 1$, 而归一化条件要求

$$\sum_{n=1}^{\infty} |C_n|^2 = \sum_{n=1}^{\infty} P_n = 1,$$

因此 $P_1 \gg P_n$, 两条曲线应非常相似。

(c) 实线为 $\psi_1(x)$, 虚线为 $\psi(x)$ 。



5

同上题。设粒子处于基态 ($n = 1$),

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}.$$

设 $t = 0$ 时刻阱宽突然变为 $2a$, 粒子波函数来不及改变, 即

$$\psi(x, 0) = \psi_1(x) = \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a}.$$

试问: 对于加宽了的无限深方势阱

$$V(x) = \begin{cases} 0, & 0 < x < 2a, \\ \infty, & x < 0, x > 2a \end{cases}$$

$\psi(x, 0)$ 是否还是能量本征态? 求测得粒子能量仍为 E_1 的概率。(答: $1/2$)

提示: 阱宽为 $2a$ 的无限深方势阱中的粒子能量本征值

$$\varepsilon_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2},$$

本征函数

$$\psi_n(x) = \sqrt{\frac{1}{a}} \sin \frac{n\pi x}{2a}.$$

$\varepsilon_2 = E_1$; $\psi(x, 0)$ 用 $\psi_n(x)$ 展开, $\psi(x, 0) = \sum_n C_n \psi_n(x)$, 求出 $|C_2|^2$ 。

解: 阱宽突变后, 粒子波函数仍为

$$\psi(x, 0) = \begin{cases} \sqrt{\frac{2}{a}} \sin \left(\frac{\pi x}{a} \right), & 0 < x < a, \\ 0, & a < x < 2a. \end{cases}$$

新的无限深方势阱 $0 < x < 2a$ 的本征函数与本征值为

$$\psi_n(x) = \sqrt{\frac{1}{a}} \sin \left(\frac{n\pi x}{2a} \right), \quad \varepsilon_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2}.$$

由于

$$\varepsilon_2 = \frac{4\pi^2\hbar^2}{8ma^2} = \frac{\pi^2\hbar^2}{2ma^2} = E_1,$$

故测得粒子能量仍为 E_1 的概率为

$$P(E_1) = |C_2|^2,$$

其中

$$\begin{aligned} C_2 &= \int_0^{2a} \psi_2(x) \psi(x, 0) dx \\ &= \int_0^a \sqrt{\frac{1}{a}} \sin\left(\frac{2\pi x}{2a}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) dx \\ &= \frac{\sqrt{2}}{a} \int_0^a \sin^2\left(\frac{\pi x}{a}\right) dx \\ &= \frac{\sqrt{2}}{a} \cdot \frac{a}{2} \\ &= \frac{1}{\sqrt{2}}. \end{aligned}$$

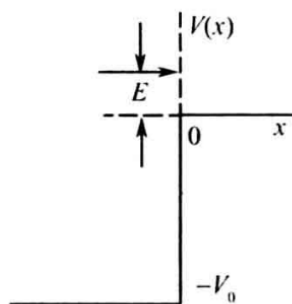
因此

$$P(E_1) = |C_2|^2 = \frac{1}{2}.$$

所以, $\psi(x, 0)$ 不是加宽后势阱的能量本征态; 测得粒子能量仍为 E_1 的概率为 $\frac{1}{2}$ 。

6

设粒子 (能量 $E > 0$) 从左入射, 碰到图 2.15 所示的势阱, 求透射系数与反射系数。



解: 设势能为

$$V(x) = \begin{cases} -V_0, & x < 0, \\ 0, & x > 0. \end{cases}$$

粒子自左向右入射，且 $E > 0$ 。

两侧波数分别为

$$k_1 = \frac{\sqrt{2m(E + V_0)}}{\hbar}, \quad k_2 = \frac{\sqrt{2mE}}{\hbar}.$$

于是

$$\psi(x) = \begin{cases} Ae^{ik_1x} + Be^{-ik_1x}, & x < 0, \\ Ce^{ik_2x}, & x > 0. \end{cases}$$

在 $x = 0$ 处，波函数与其一阶导数连续：

$$A + B = C,$$

$$ik_1(A - B) = ik_2C.$$

解得：

$$\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}.$$

$$\frac{C}{A} = 1 + \frac{k_1 - k_2}{k_1 + k_2} = \frac{2k_1}{k_1 + k_2}.$$

故反射系数为

$$R = \left| \frac{B}{A} \right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2.$$

透射系数为

$$T = \frac{j_t}{j_i} = \frac{k_2}{k_1} \left| \frac{C}{A} \right|^2 = \frac{k_2}{k_1} \left(\frac{2k_1}{k_1 + k_2} \right)^2 = \frac{4k_1k_2}{(k_1 + k_2)^2}.$$

其中

$$k_1 = \frac{\sqrt{2m(E + V_0)}}{\hbar}, \quad k_2 = \frac{\sqrt{2mE}}{\hbar}.$$

7

利用 Hermite 多项式的递推关系（附录 A3，式 (13)），证明谐振子波函数满足下列关系

$$x\psi_n(x) = \frac{1}{\alpha} \left[\sqrt{\frac{n}{2}} \psi_{n-1}(x) + \sqrt{\frac{n+1}{2}} \psi_{n+1}(x) \right]$$

$$x^2\psi_n(x) = \frac{1}{2\alpha^2} \left[\sqrt{n(n-1)} \psi_{n-2}(x) + (2n+1)\psi_n(x) + \sqrt{(n+1)(n+2)} \psi_{n+2}(x) \right].$$

并由此证明，在 ψ_n 态下， $\bar{x} = 0$ ， $\bar{V} = E_n/2$ 。

证明： 谐振子归一化波函数可写为

$$\psi_n(x) = N_n H_n(\xi) e^{-\xi^2/2}, \quad \xi = \alpha x,$$

其中

$$N_n = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{1/2}.$$

Hermite 多项式的递推关系为

$$2\xi H_n(\xi) = 2nH_{n-1}(\xi) + H_{n+1}(\xi).$$

于是

$$x\psi_n(x) = \frac{\xi}{\alpha}\psi_n(x) = \frac{N_n}{2\alpha} [2nH_{n-1}(\xi) + H_{n+1}(\xi)] e^{-\xi^2/2}.$$

又由于

$$\psi_{n-1}(x) = N_{n-1}H_{n-1}(\xi)e^{-\xi^2/2}, \quad \psi_{n+1}(x) = N_{n+1}H_{n+1}(\xi)e^{-\xi^2/2},$$

并且

$$\frac{nN_n}{N_{n-1}} = \sqrt{\frac{n}{2}}, \quad \frac{N_n}{2N_{n+1}} = \sqrt{\frac{n+1}{2}},$$

故

$$x\psi_n(x) = \frac{1}{\alpha} \left[\sqrt{\frac{n}{2}} \psi_{n-1}(x) + \sqrt{\frac{n+1}{2}} \psi_{n+1}(x) \right].$$

再对上式再乘一个 x , 得

$$x^2\psi_n(x) = \frac{1}{\alpha} \sqrt{\frac{n}{2}} x\psi_{n-1}(x) + \frac{1}{\alpha} \sqrt{\frac{n+1}{2}} x\psi_{n+1}(x).$$

分别代入前一结果:

$$\begin{aligned} x\psi_{n-1}(x) &= \frac{1}{\alpha} \left[\sqrt{\frac{n-1}{2}} \psi_{n-2}(x) + \sqrt{\frac{n}{2}} \psi_n(x) \right], \\ x\psi_{n+1}(x) &= \frac{1}{\alpha} \left[\sqrt{\frac{n+1}{2}} \psi_n(x) + \sqrt{\frac{n+2}{2}} \psi_{n+2}(x) \right]. \end{aligned}$$

因此

$$x^2\psi_n(x) = \frac{1}{2\alpha^2} \left[\sqrt{n(n-1)} \psi_{n-2}(x) + (2n+1)\psi_n(x) + \sqrt{(n+1)(n+2)} \psi_{n+2}(x) \right].$$

对第一式左乘 $\psi_n^*(x)$ 并积分, 得

$$\bar{x} = \langle x \rangle = \frac{1}{\alpha} \left[\sqrt{\frac{n}{2}} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n-1}(x) dx + \sqrt{\frac{n+1}{2}} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n+1}(x) dx \right] = 0.$$

再对第二式左乘 $\psi_n^*(x)$ 并积分, 得

$$\begin{aligned} \langle x^2 \rangle &= \frac{1}{2\alpha^2} \left[\sqrt{n(n-1)} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n-2}(x) dx \right. \\ &\quad + (2n+1) \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_n(x) dx \\ &\quad \left. + \sqrt{(n+1)(n+2)} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n+2}(x) dx \right] \\ &= \frac{2n+1}{2\alpha^2}. \end{aligned}$$

谐振子势能为

$$V(x) = \frac{1}{2}m\omega^2 x^2,$$

故

$$\bar{V} = \frac{1}{2}m\omega^2 \langle x^2 \rangle = \frac{m\omega^2}{2} \cdot \frac{2n+1}{2\alpha^2} = \frac{m\omega^2}{2} \cdot \frac{2n+1}{2} \cdot \frac{\hbar}{m\omega} = \frac{2n+1}{4}\hbar\omega.$$

而

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega = \frac{2n+1}{2}\hbar\omega,$$

所以

$$\bar{V} = \frac{E_n}{2}.$$

故证得

$$\bar{x} = 0, \quad \bar{V} = \frac{E_n}{2}.$$

8

同上题, 利用 Hermite 多项式的求导公式 (附录 A3, 式 (14)), 证明

$$\frac{d}{dx}\psi_n(x) = \alpha \left[\sqrt{\frac{n}{2}} \psi_{n-1} - \sqrt{\frac{n+1}{2}} \psi_{n+1} \right]$$

$$\frac{d^2}{dx^2}\psi_n(x) = \frac{\alpha^2}{2} \left[\sqrt{n(n-1)} \psi_{n-2} - (2n+1)\psi_n + \sqrt{(n+1)(n+2)} \psi_{n+2} \right].$$

并由此证明, 在 ψ_n 态下, $\bar{p} = 0$, $\bar{T} = \bar{p}^2/2m = E_n/2$ 。

证明: 谐振子归一化波函数为

$$\psi_n(x) = N_n H_n(\xi) e^{-\xi^2/2}, \quad \xi = \alpha x,$$

其中

$$N_n = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{1/2}.$$

Hermite 多项式的求导公式为

$$\frac{dH_n}{d\xi} = 2nH_{n-1}(\xi),$$

因此:

$$\begin{aligned}
 \frac{d\psi_n}{dx} &= \alpha N_n \frac{d}{d\xi} [H_n(\xi) e^{-\xi^2/2}] \\
 &= \alpha N_n \left[\frac{dH_n}{d\xi} - \xi H_n(\xi) \right] e^{-\xi^2/2} \\
 &= \alpha N_n [2nH_{n-1}(\xi) - \xi H_n(\xi)] e^{-\xi^2/2} \\
 &= \alpha N_n \left[2nH_{n-1}(\xi) - \frac{1}{2} (2nH_{n-1}(\xi) + H_{n+1}(\xi)) \right] e^{-\xi^2/2} \\
 &= \alpha N_n \left[nH_{n-1}(\xi) - \frac{1}{2} H_{n+1}(\xi) \right] e^{-\xi^2/2} \\
 &= \alpha \left[\frac{nN_n}{N_{n-1}} \psi_{n-1}(x) - \frac{N_n}{2N_{n+1}} \psi_{n+1}(x) \right] \\
 &= \alpha \left[\sqrt{\frac{n}{2}} \psi_{n-1}(x) - \sqrt{\frac{n+1}{2}} \psi_{n+1}(x) \right].
 \end{aligned}$$

再对上式求导,

$$\frac{d^2\psi_n}{dx^2} = \alpha \sqrt{\frac{n}{2}} \frac{d\psi_{n-1}}{dx} - \alpha \sqrt{\frac{n+1}{2}} \frac{d\psi_{n+1}}{dx}.$$

这里

$$\begin{aligned}
 \frac{d\psi_{n-1}}{dx} &= \alpha \left[\sqrt{\frac{n-1}{2}} \psi_{n-2}(x) - \sqrt{\frac{n}{2}} \psi_n(x) \right], \\
 \frac{d\psi_{n+1}}{dx} &= \alpha \left[\sqrt{\frac{n+1}{2}} \psi_n(x) - \sqrt{\frac{n+2}{2}} \psi_{n+2}(x) \right].
 \end{aligned}$$

故

$$\frac{d^2\psi_n}{dx^2} = \frac{\alpha^2}{2} \left[\sqrt{n(n-1)} \psi_{n-2}(x) - (2n+1) \psi_n(x) + \sqrt{(n+1)(n+2)} \psi_{n+2}(x) \right].$$

动量算符为

$$\hat{p} = -i\hbar \frac{d}{dx}.$$

于是

$$\bar{p} = \int_{-\infty}^{+\infty} \psi_n^*(x) \hat{p} \psi_n(x) dx = -i\hbar \int_{-\infty}^{+\infty} \psi_n^*(x) \frac{d\psi_n}{dx} dx.$$

代入上式,

$$\bar{p} = -i\hbar \alpha \left[\sqrt{\frac{n}{2}} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n-1}(x) dx - \sqrt{\frac{n+1}{2}} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n+1}(x) dx \right] = 0.$$

又

$$\hat{p}^2 = -\hbar^2 \frac{d^2}{dx^2},$$

从而

$$\begin{aligned}
 \overline{p^2} &= \int_{-\infty}^{+\infty} \psi_n^*(x) \hat{p}^2 \psi_n(x) dx = -\hbar^2 \int_{-\infty}^{+\infty} \psi_n^*(x) \frac{d^2 \psi_n}{dx^2} dx \\
 &= -\frac{\hbar^2 \alpha^2}{2} \left[\sqrt{n(n-1)} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n-2}(x) dx \right. \\
 &\quad \left. - (2n+1) \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_n(x) dx \right. \\
 &\quad \left. + \sqrt{(n+1)(n+2)} \int_{-\infty}^{+\infty} \psi_n^*(x) \psi_{n+2}(x) dx \right] \\
 &= \frac{\hbar^2 \alpha^2}{2} (2n+1) \\
 &= \frac{\hbar m \omega}{2} (2n+1).
 \end{aligned}$$

因此

$$\overline{T} = \frac{\overline{p^2}}{2m} = \frac{\hbar \omega}{4} (2n+1).$$

而

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega = \frac{2n+1}{2} \hbar \omega,$$

故

$$\overline{T} = \frac{E_n}{2}.$$

因此

$$\overline{p} = 0, \quad \overline{T} = \frac{\overline{p^2}}{2m} = \frac{E_n}{2}.$$

9

谐振子处于 ψ_n 态下, 计算

$$\Delta x = \left[\overline{(x - \bar{x})^2} \right]^{1/2}, \quad \Delta p = \left[\overline{(p - \bar{p})^2} \right]^{1/2}, \quad \Delta x \Delta p = ?$$

解: 由前两题结果, 在 ψ_n 态下,

$$\bar{x} = 0, \quad \bar{p} = 0,$$

$$\overline{x^2} = \frac{2n+1}{2\alpha^2}, \quad \overline{p^2} = \frac{\hbar^2 \alpha^2}{2} (2n+1).$$

因此

$$\Delta x = \left[\overline{(x - \bar{x})^2} \right]^{1/2} = \left(\overline{x^2} - \bar{x}^2 \right)^{1/2} = \sqrt{\overline{x^2}} = \frac{1}{\alpha} \sqrt{\frac{2n+1}{2}}.$$

同理

$$\Delta p = \left[\overline{(p - \bar{p})^2} \right]^{1/2} = \left(\overline{p^2} - \bar{p}^2 \right)^{1/2} = \sqrt{\overline{p^2}} = \hbar \alpha \sqrt{\frac{2n+1}{2}}.$$

故

$$\Delta x \Delta p = \frac{1}{\alpha} \sqrt{\frac{2n+1}{2}} \cdot \hbar \alpha \sqrt{\frac{2n+1}{2}} = \frac{2n+1}{2} \hbar.$$

10

荷电 q 的谐振子, 受到外电场 $\mathcal{E}$ 的作用,

$$V(x) = \frac{1}{2} m \omega^2 x^2 - q \mathcal{E} x$$

求能量本征值和本征函数。(提示: 对 $V(x)$ 进行配方, 相当于谐振子势的平衡点不在 $x = 0$ 点。)

解: 势能可配方为

$$V(x) = \frac{1}{2} m \omega^2 x^2 - q \mathcal{E} x = \frac{1}{2} m \omega^2 \left(x - \frac{q \mathcal{E}}{m \omega^2} \right)^2 - \frac{q^2 \mathcal{E}^2}{2 m \omega^2}.$$

记

$$x_0 = \frac{q \mathcal{E}}{m \omega^2},$$

则

$$V(x) = \frac{1}{2} m \omega^2 (x - x_0)^2 - \frac{q^2 \mathcal{E}^2}{2 m \omega^2}.$$

定态薛定谔方程为

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 (x - x_0)^2 - \frac{q^2 \mathcal{E}^2}{2 m \omega^2} \right] \psi(x) = E \psi(x).$$

令

$$\xi = x - x_0,$$

则方程化为

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{d\xi^2} + \frac{1}{2} m \omega^2 \xi^2 \right] \psi(\xi) = \left(E + \frac{q^2 \mathcal{E}^2}{2 m \omega^2} \right) \psi(\xi).$$

这就是以 $\xi = 0$ 为平衡位置的标准谐振子方程, 因此

$$E + \frac{q^2 \mathcal{E}^2}{2 m \omega^2} = \left(n + \frac{1}{2} \right) \hbar \omega, \quad n = 0, 1, 2, \dots$$

故能量本征值为

$$E_n = \left(n + \frac{1}{2} \right) \hbar \omega - \frac{q^2 \mathcal{E}^2}{2 m \omega^2}, \quad n = 0, 1, 2, \dots$$

本征函数就是谐振子本征函数整体平移 x_0 后的结果。记

$$\alpha = \sqrt{\frac{m\omega}{\hbar}},$$

则本征函数为

$$\psi_n(x) = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{1/2} H_n \left[\alpha \left(x - \frac{q\mathcal{E}}{m\omega^2} \right) \right] \exp \left[-\frac{\alpha^2}{2} \left(x - \frac{q\mathcal{E}}{m\omega^2} \right)^2 \right].$$

11

设粒子在下列势阱中运动，求粒子能级，

$$V(x) = \begin{cases} \infty, & x < 0, \\ \frac{1}{2}m\omega^2 x^2, & x > 0. \end{cases}$$

解：在 $x < 0$ 区域，势能 $V(x) = \infty$ ，故波函数必须满足

$$\psi(x) = 0, \quad x < 0,$$

并且在边界处有

$$\psi(0) = 0.$$

在 $x > 0$ 区域，势能为

$$V(x) = \frac{1}{2}m\omega^2 x^2,$$

故定态薛定谔方程就是标准一维谐振子的方程。其本征值为

$$E_n = \left(n + \frac{1}{2} \right) \hbar\omega, \quad n = 0, 1, 2, \dots$$

本征函数为

$$\psi_n(x) = N_n H_n(\alpha x) e^{-\alpha^2 x^2/2}, \quad \alpha = \sqrt{\frac{m\omega}{\hbar}}.$$

由于这里还要求

$$\psi(0) = 0,$$

所以只能取在 $x = 0$ 处为零的谐振子本征函数。而 Hermite 多项式满足

$$H_n(0) = 0 \iff n \text{ 为奇数},$$

故该势阱中的允许能级为

$$E_n = \left(n + \frac{1}{2} \right) \hbar\omega, \quad n = 1, 3, 5, \dots$$

令

$$n = 2k + 1,$$

则

$$E_k = \left(2k + \frac{3}{2} \right) \hbar\omega, \quad k = 0, 1, 2, \dots$$

12

一维无限深方势阱中的粒子, 设初始时刻 ($t = 0$) 处于

$$\psi(x, 0) = \frac{\psi_1(x) + \psi_2(x)}{\sqrt{2}},$$

$\psi_1(x)$ 与 $\psi_2(x)$ 分别为基态和第一激发态。求

(a) $\psi(x, t)$, $\rho(x, t) = \psi^*(x, t)\psi(x, t)$;

(b) 能量平均值 $\overline{H}$;

(c) 能量平方平均值 $\overline{H^2}$;

(d) 能量的涨落

$$\Delta E = \left[\overline{(H - \overline{H})^2} \right]^{1/2};$$

(e) 体系的特征时间 Δt (分析 $\rho(x, t)$ 随时间变化的周期)。计算 $\Delta E \cdot \Delta t$ 。

解: 一维无限深方势阱的能量本征态满足

$$\hat{H}\psi_n(x) = E_n\psi_n(x), \quad E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}.$$

因此

$$E_1 = \frac{\pi^2\hbar^2}{2ma^2}, \quad E_2 = \frac{4\pi^2\hbar^2}{2ma^2} = 4E_1.$$

(a) 由定态叠加的时间演化,

$$\psi(x, t) = \frac{1}{\sqrt{2}} [\psi_1(x)e^{-iE_1t/\hbar} + \psi_2(x)e^{-iE_2t/\hbar}].$$

于是

$$\psi^*(x, t) = \frac{1}{\sqrt{2}} [\psi_1(x)e^{iE_1t/\hbar} + \psi_2(x)e^{iE_2t/\hbar}].$$

故概率密度为

$$\begin{aligned} \rho(x, t) &= \psi^*(x, t)\psi(x, t). \\ \rho(x, t) &= \frac{1}{2} \left[\psi_1^2(x) + \psi_2^2(x) + 2\psi_1(x)\psi_2(x) \cos\left(\frac{E_2 - E_1}{\hbar}t\right) \right]. \end{aligned}$$

(b) 由于 ψ_1, ψ_2 是哈密顿量本征态且彼此正交,

$$\overline{H} = \langle H \rangle = \langle \psi | \hat{H} | \psi \rangle = \frac{1}{2} (E_1 + E_2) = \frac{5\pi^2\hbar^2}{4ma^2}.$$

(c) 同理,

$$\overline{H^2} = \langle H^2 \rangle = \frac{1}{2} (E_1^2 + E_2^2) = \frac{17}{2} E_1^2.$$

(d) 能量涨落为

$$\begin{aligned}\Delta E &= \left[\overline{(H - \bar{H})^2} \right]^{1/2} = \left(\overline{H^2} - \bar{H}^2 \right)^{1/2} \\ &= \left[\frac{1}{2} (E_1^2 + E_2^2) - \frac{1}{4} (E_1 + E_2)^2 \right]^{1/2} \\ &= \frac{1}{2} |E_2 - E_1| = 3E_1 = \frac{3\pi^2 \hbar^2}{4ma^2}.\end{aligned}$$

(e) $\rho(x, t)$ 随时间的变化只出现在因子

$$\cos\left(\frac{E_2 - E_1}{\hbar}t\right)$$

中，因此其周期为

$$\Delta t = \frac{2\pi\hbar}{E_2 - E_1}.$$

由于

$$E_2 - E_1 = 3E_1 = \frac{3\pi^2 \hbar^2}{2ma^2},$$

故

$$\Delta t = \frac{2\pi\hbar}{3E_1} = \frac{4ma^2}{3\pi\hbar}.$$

因此

$$\Delta E \cdot \Delta t = \frac{1}{2} (E_2 - E_1) \cdot \frac{2\pi\hbar}{E_2 - E_1} = \pi\hbar.$$